
T
N
Binary Mask
Complete Trajectory
Input Process
GSM Module
T
N
Agent embeddings
Temporal Decay
Spatial agent embeddings
Bidirectional Temporal Mamba Block
× L
Trajectory decoding with decoupled queries
Mode CrossAttn
Mode SelfAttn
(a) Mode Localization Module
State CrossAttn
State BiMamba
(b) State Consistency Module
$\mathcal{L}_{ts}$
$\mathcal{L}_m$
(B, K, T_s, C)
Hybrid CrossAttn (B, KT_s, C)
Hybrid SelfAttn (B, KT_s, C)
Mode SelfAttn (BT_s, K, C)
State BiMamba (BK, T_s, C)
(c) Hybrid Coupling Module
Trajectories
$\mathcal{L}_{reg}$
Probabilities
$\mathcal{L}_{cls}$